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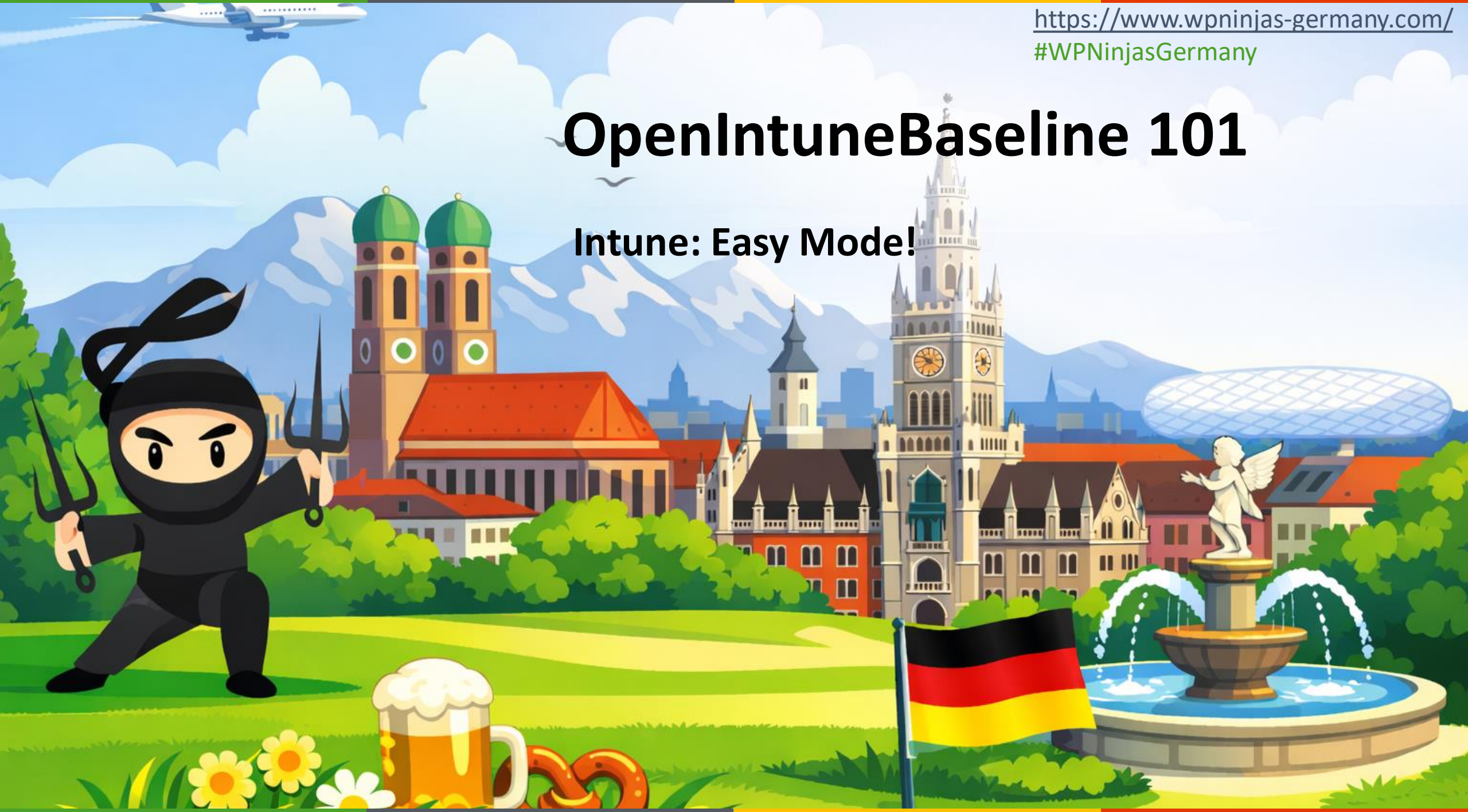
OpenIntuneBaseline 101

Intune: Easy Mode!



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SOFTWARE CENTRAL

Silver Sponsors

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Community Sponsor

Microsoft

Bronze Sponsors

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aconitas®
EPIC FUSION BRING IT ALL TOGETHER
WORKSIMPLE EINFACH ARBEITEN

About James

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Focus

Intune and Intune Accessories

From

Brighton(ish), United Kingdom

My Blog

skiptotheendpoint.co.uk

Certifications

- Intune & Windows MVP
- Neurodivergent Loudmouth

Hobbies

- Shouting at Security Teams
- Grumbling about poor UI/UX

Contact

- SkipToTheEndpoint
- @SkipToEndpoint

How It All Began

A trip down memory lane…

- I h
- Eve
- I d
- … T

Files

main

Go to file

- IntuneBaseline
  - AppProtection
  - Applications
  - AssignmentFilters
  - AutoPilot
  - CompliancePolicies
  - ConditionalAccess
  - DeviceConfiguration
  - EndpointSecurity
  - EnrollmentRestrictions
  - EnrollmentStatusPage
  - FeatureUpdates
  - Groups
  - PowerShellScripts
  - SettingsCatalog
    - Defender ASR - ASR (Audi...
    - Defender ASR - ASR (Bloc...
    - Defender Antivirus Exclusi...
    - Defender Antivirus_f6046...
    - Defender EDR_2e054db2-...
    - Defender Firewall_450b0e...
    - Defender Security Experie...
    - Desktop Analytics_d7629...
    - Device Security Hardenin...
    - Google Chrome - Security...
    - Google Chrome - User Ex...
    - Local Device Security_f8d...
    - Lock Screen Experience_5...

projects / Intune Baseline / IntuneBaseline / SettingsCatalog /

Add file

SkipToTheEndpoint Add files via upload

37a6047 · 4 years ago History

Name	Last commit message	Last commit date
..		
Defender ASR - ASR (Audit)_a4dad0d4-db22-4269-ab5f-5d4d86edd519.json	Upload	4 years ago
Defender ASR - ASR (Block)_fe3c98a2-da0b-46b8-a4fd-aeb208eaf0d2.json	Upload	4 years ago
Defender Antivirus Exclusions_c5499e60-3ef1-4ca3-8dc9-c88ffa8306a5.json	Upload	4 years ago
Defender Antivirus_f60468ee-5a09-465b-84d3-34cb84f07206.json	Upload	4 years ago
Defender EDR_2e054db2-89c1-4ce0-9ddc-bd7c22c1dbfa.json	Upload	4 years ago
Defender Firewall_450b0e2b-915b-4530-9773-326a7bf69e11.json	Add files via upload	4 years ago
Defender Security Experience_86e85c5d-18f3-406a-a50d-8200df3bed88.json	Upload	4 years ago
Desktop Analytics_d76293d4-5a22-4e9c-86d8-d1baccdbd6b6.json	Upload	4 years ago
Device Security Hardening_ef0a412e-529a-4226-b3d9-70d24c4506c9.json	Upload	4 years ago
Google Chrome - Security (Device)_3e5ac1a6-ef8d-4630-a3a0-17845df868ab.json	Upload	4 years ago
Google Chrome - User Experience & Extensions (User)_49bca29d-402b-4b05-91ea-c0449844...	Upload	4 years ago
Local Device Security_f8d2051f-065c-4e37-9654-dfc1a3fd1b2b.json	Upload	4 years ago
Lock Screen Experience_53162abb-2963-4f1c-aa20-b5bbf88b6611.json	Upload	4 years ago
MS Store and Accounts_5314ed31-7036-4806-a324-211d6b9af5f2.json	Upload	4 years ago
Microsoft Edge - Security (Device)_c3e130d5-63e2-4d75-a7c9-20abac4e4aed.json	Upload	4 years ago
Microsoft Edge - User Experience & Extensions (User)_7cb5136d-c945-45bf-b2d4-fafab3458e...	Upload	4 years ago
Office - Outlook (User)_1070c3ee-8072-4d46-a935-82e845bf464f.json	Upload	4 years ago
Office - Update Settings (Device)_bac83fee-fbff-4848-bec3-bfb2bd81ead0.json	Upload	4 years ago
OneDrive (Device)_17079ac2-7abe-4b8c-b0a7-d21e18242d54.json	Upload	4 years ago
OneDrive (User)_63841a0a-af64-4a8d-8ab5-b920ae617f0d.json	Upload	4 years ago
Power & Device Lock_86638ec3-eb07-4520-a4e3-0c87b3aa5fbc.json	Upload	4 years ago
Preferred AAD Tenant Domain_83f4b35a-7a50-4239-becb-4efd78aeb12b.json	Upload	4 years ago
Privacy_cec56338-8c5e-45ad-9f8c-b0cdfe22fefe.json	Upload	4 years ago
Self-Service Password Reset_c0ee0ec6-8f3a-4b7c-9622-766f3156c3e5.json	Upload	4 years ago
Storage Sense_90ae8980-6b53-4c6b-acdd-bc335e312601.json	Upload	4 years ago
Wi-Fi, Cellular & Connectivity_535c146a-5243-417a-ad85-5aa145fdde25.json	Upload	4 years ago
Windows Update Compliance Telemetry (Win Ent)_0a9564cb-6383-4287-af6f-2fba056fd6aa.js...	Upload	4 years ago
Windows Update Delivery Optimisation_02105da4-192a-4820-8ef2-855390b5ba49.json	Upload	4 years ago

rmany.com/

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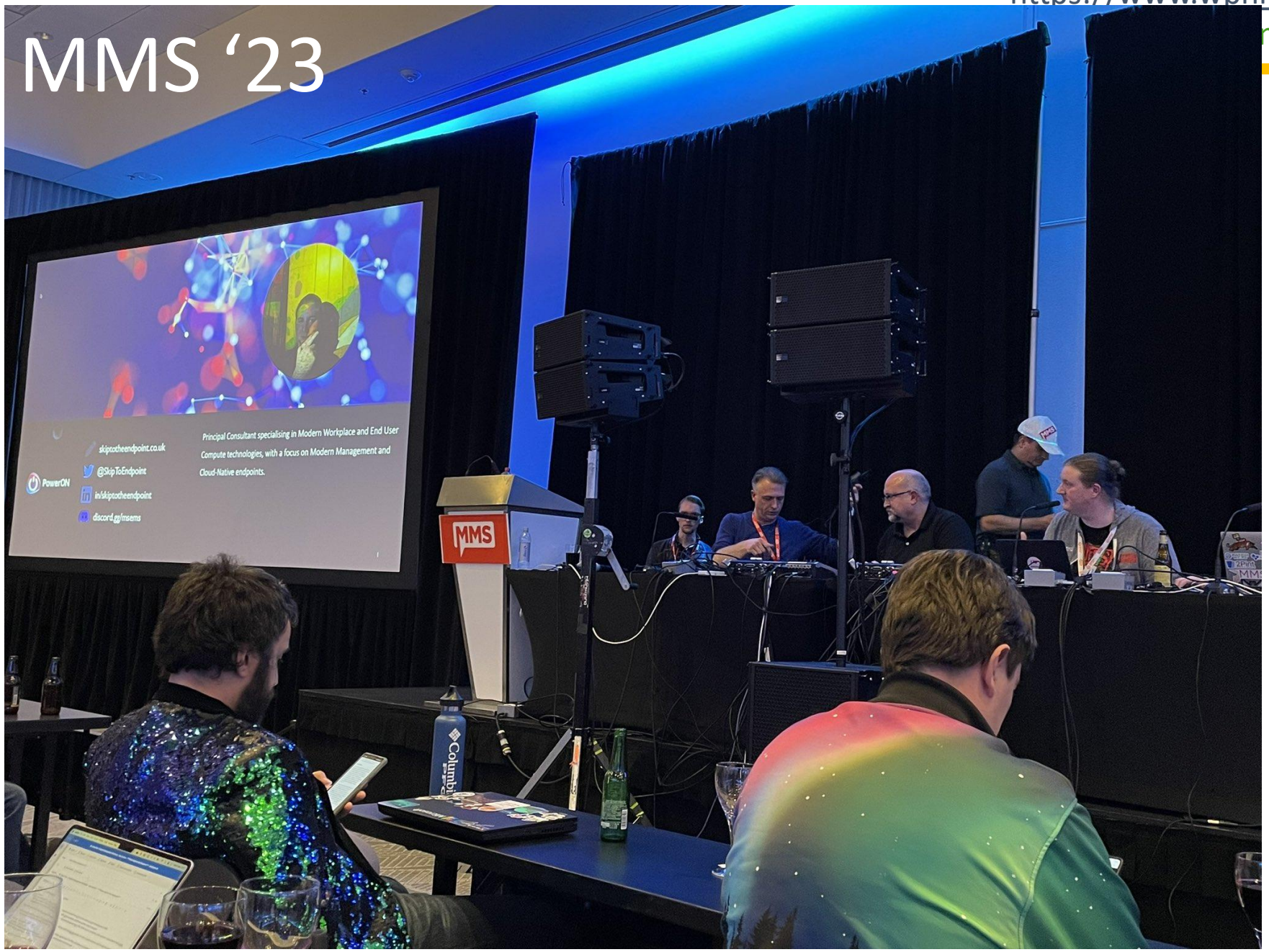
SkipToTheEndpoint Initial baseline commit — 0ba7175 · 4 years ago — 2 Commits

AppProtection	Initial baseline commit	4 years ago
AutoPilot	Initial baseline commit	4 years ago
CompliancePolicies	Initial baseline commit	4 years ago
ConditionalAccess	Initial baseline commit	4 years ago
DeviceConfiguration	Initial baseline commit	4 years ago
EndpointSecurity	Initial baseline commit	4 years ago
EnrollmentRestrictions	Initial baseline commit	4 years ago
EnrollmentStatusPage	Initial baseline commit	4 years ago
FeatureUpdates	Initial baseline commit	4 years ago
Groups	Initial baseline commit	4 years ago
SettingsCatalog	Initial baseline commit	4 years ago
UpdatePolicies	Initial baseline commit	4 years ago
LICENSE	Initial commit	4 years ago
MigrationTable.json	Initial baseline commit	4 years ago
README.md	Initial baseline commit	4 years ago

SkipToTheEndpoint OIB v1.3.1 — 729427b · 4 years ago — 8 Commits

AppProtection	OIB v1.2	4 years ago
AssignmentFilters	OIB v1.2	4 years ago
AutoPilot	OIB v1.2	4 years ago
CompliancePolicies	OIB v1.2	4 years ago
ConditionalAccess	OIB v1.2	4 years ago
DeviceConfiguration	OIB v1.2	4 years ago
EndpointSecurity	OIB v1.2	4 years ago
EnrollmentRestrictions	OIB v1.2	4 years ago
FeatureUpdates	OIB v1.2	4 years ago
Groups	OIB v1.2	4 years ago
SettingsCatalog	OIB v1.3.1	4 years ago
UpdatePolicies	OIB v1.2	4 years ago
CHANGELOG.md	OIB v1.3.1	4 years ago
LICENSE	Initial commit	4 years ago
MigrationTable.json	OIB v1.2	4 years ago
README.md	OIB v1.2.1	4 years ago
SETTINGS.md	OIB v1.2	4 years ago

MMS ‘23


skiptotheendpoint.co.uk
@SkipToEndpoint
in/skiptotheendpoint
discord.gg/msems
PowerON
Principal Consultant specialising in Modern Workplace and End User Compute technologies, with a focus on Modern Management and Cloud-Native endpoints.
MMS

So What Is It?

And why should we trust you?

What the OpenIntuneBaseline isn't!

- CIS Benchmark
- Your mum
- Meant for Hybrid Joined or Shared Devices (or Windows 10)
- An excuse to not understand how Intune works

Here Be Dragons!

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Just because I did all the hard work doesn’t mean you don’t need to understand what it’s doing, and why.

Recent Developments

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CIS Microsoft Intune for Windows 11 Benchmark v4.0.0

CIS Microsoft Windows Workstation Benchmark Baseline vNext

CIS Microsoft Intune for Microsoft Windows - Intune Windows 10/11 v4.0.0

Published 6 days ago on Apr 25th 2025 **Release Type:** Planned Update

Overview

This document provides prescriptive guidance for establishing a secure configuration posture for Microsoft Windows Operating Systems (OS). This secure configuration guide is based on Windows 11 and is intended for all releases of the Windows 11 operating system, including older versions. This secure configuration guide was tested against

Ensure that the latest vers
CIS on-prem Windows OS
this benchmark is based o
publication.

Note: If **Windows Autopi**
groups. This ensures the
interruptions. For more information visit: Windows Autopilot - Policy Conflicts.

To obtain the latest version of this secure configuration guide, please visit the CIS Website or the CIS WorkBench Community. If you have questions, comments, or have identified ways to improve this guide, please write to us at feedback@cisecurity.org.

Intended Audience

The Microsoft Intune Windows Benchmarks are written for **MDM-joined** systems using Microsoft Intune device configuration profiles only. This benchmark is **not** intended for use on standalone or workgroup systems, systems joined to and receive policies from Active Directory, or systems created, maintained, or used in other Cloud offerings. This benchmark covers supported endpoint states for **Entra Hybrid Joined** and **Entra Joined** systems that receive policies from **Microsoft Intune** using **Configuration Profiles** only.

Acknowledgements

The Center for Internet Security extends special recognition and thanks to the WorkBench Intune Community for their collaboration developing the configuration recommendations contained in this document.

Contributors

Haemish Edgerton, Hardeep Mehrotara, Kevin Zhang, Jennifer Jarose, Phil White, Matthew Woods, Caleb Eifert, Phil Chatham, Uzoma Ifeakanwa, JJ Milner, Cody White, Martin Himken, Rex Farabee, James Robinson, Daniel Wahlgren, Steven Los, William Ferguson, Nick Benton, Justin Hall, Johannes Kristjansson, Jennifer Jarose

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What is the OpenIntuneBaseline?

- A learning resource
- A source of inspiration
- More than just security hardening
- Trusted and used in thousands of tenants

Stats for Nerds

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Currently:

- 1.2k+ Stars
- 230+ Forks

Monthly stats (average):

- 6000+ unique visitors
  - Google, Reddit, LinkedIn & Teams!
- ~500 unique clones

Yeah but you're just a guy with a lot of opinions…

- Overseen deployment into ~50 tenants – 10 to 25k users.
- Utilised by large consulting firms across Europe.
- I <u>really</u> care about this stuff.

Demo: Yeet to Prod!

Import Demo

- Use IntuneManagement
- Use OIB Deployer

Alright… Now What?

The curse of Intune assignments

(User) vs. (Device)

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- [ ] Ads setting for sites with intrusive ads (Device)
- [ ] Allow Option To Show This PC (User)

DisablePasswordReveal

Expand table

Scope	Editions	Applicable OS
✅ Device ✅ User	✅ Pro ✅ Enterprise ✅ Education ✅ IoT Enterprise / IoT Enterprise LTSC	✅ Windows 10, version 1703 [10.0.15063] and later

HKCU

User — Copy

```
./User/Vendor/MSFT/Policy/Config/CredentialsUI/DisablePasswordReveal
```

HKLM

Device — Copy

```
./Device/Vendor/MSFT/Policy/Config/CredentialsUI/DisablePasswordReveal
```

This is Fine…

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☐ Enable Virtualization Based Security

EnableVirtualizationBasedSecurity

Expand table

Scope	Editions	Applicable OS
✅ Device ❌ User	✅ Pro ✅ Enterprise ✅ Education ✅ IoT Enterprise / IoT Enterprise LTSC	✅ Windows 10, version 1709 [10.0.16299] and later

Device — Copy

```
./Device/Vendor/MSFT/Policy/Config/DeviceGuard/EnableVirtualizationBasedSecurity
```

RequirePlatformSecurityFeatures

Expand table

Scope	Editions	Applicable OS
✅ Device ❌ User	❌ Pro ✅ Enterprise ✅ Education ✅ IoT Enterprise / IoT Enterprise LTSC	✅ Windows 10, version 1709 [10.0.16299] and later

Device — Copy

```
./Device/Vendor/MSFT/Policy/Config/DeviceGuard/RequirePlatformSecurityFeatures
```

ConfigureLsaProtectedProcess

Expand table

Scope	Editions	Applicable OS
✅ Device ❌ User	✅ Pro ✅ Enterprise ✅ Education ✅ IoT Enterprise / IoT Enterprise LTSC	✅ Windows 11, version 22H2 [10.0.22621] and later

Device — Copy

```
./Device/Vendor/MSFT/Policy/Config/LocalSecurityAuthority/ConfigureLsaProtectedProcess
```

Assignment Behaviour

- *If a device scoped policy is assigned to a user, once that user signs in and an Intune sync occurs, the device scope settings apply to all users on the device.*
- *If a user scope policy is assigned to a device, all users on that device have that setting applied. This behavior is like a loopback set to merge.*

Create a policy using settings catalog in Microsoft Intune | Microsoft Learn

Assignment Behaviour

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Home > Devices | Windows > Windows | Configuration >

Win - OIB - SC - Device Security - U - Device Guard, Credential Guard and HVCI - v3

Device configuration profile

Delete

Configuration settings Edit

Device Guard

Configure System Guard Launch	Unmanaged Enables Secure Launch if supported by hardware
Credential Guard	(Enabled without lock) Turns on Credential Guard without UEFI lock.
Enable Virtualization Based Security	enable virtualization based security.
Machine Identity Isolation	(Disabled) Machine password is only LSASS-bound and stored in $MACHINE.ACC registry key.
Require Platform Security Features	Turns on VBS with Secure Boot and direct memory access (DMA). DMA requires hardware support.

Local Security Authority

Configure Lsa Protected Process	Enabled without UEFI lock. LSA will run as protected process and this configuration is not UEFI locked.

Virtualization Based Technology

Hypervisor Enforced Code Integrity	(Enabled without lock) Turns on Hypervisor-Protected Code Integrity without UEFI lock.
Require UEFI Memory Attributes Table	Require UEFI Memory Attributes Table

Computer\HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Provisioning\SyncML\RebootRequiredURIs

- Provisioning
  - Agent
  - AutopilotPolicy
  - AutopilotPolicyCache
  - AutopilotPolicyManagerOrc
  - AutopilotSettings
  - AutopilotTpmEnhancedLog
  - CSPs
  - Diagnostics
  - DMSessionActions
  - Enrollments
  - Evaluator
  - FirstBootRun
  - HardwareInfo
  - LogonTaskCompleted
  - NodeCache
  - OMADM
  - PackageInfo
  - PackageLocations
  - Plugin
  - Results
  - ServerConfig
  - Sessions
  - SyncML
    - NormalReplaceURIs
    - RebootRequiredURIs
  - TrustedProvisioners
- PushRouter
- QuickAssist
- RADAR
- Ras
- RAS AutoDial
- RcsPresence
- Reliability Analysis
- Remediation
- RemovalTools
- RendezvousApps

Name	Type
(Default)	REG_SZ
./Device/Vendor/MSFT/Accounts/Domain/ComputerName	REG_SZ
./Device/Vendor/MSFT/Policy/Config/Connectivity/AllowUSBConnection	REG_SZ
./Device/Vendor/MSFT/Policy/Config/DeviceGuard/ConfigureSystemGuardLaunch	REG_SZ
./Device/Vendor/MSFT/Policy/Config/DeviceGuard/EnableVirtualizationBasedSecurity	REG_SZ
./Device/Vendor/MSFT/Policy/Config/DeviceGuard/LsaCfgFlags	REG_SZ
./Device/Vendor/MSFT/Policy/Config/DeviceGuard/RequirePlatformSecurityFeatures	REG_SZ
./Device/Vendor/MSFT/Policy/Config/DmaGuard/DeviceEnumerationPolicy	REG_SZ
./Device/Vendor/MSFT/Policy/Config/ExploitGuard/ExploitProtectionSettings	REG_SZ
./Device/Vendor/MSFT/Policy/Config/MixedReality/HeadTrackingMode	REG_SZ
./Device/Vendor/MSFT/Policy/Config/Notifications/DisallowCloudNotification	REG_SZ
./Device/Vendor/MSFT/Policy/Config/Notifications/DisallowTileNotification	REG_SZ
./Device/Vendor/MSFT/Policy/Config/Notifications/WnsEndpoint	REG_SZ
./Device/Vendor/MSFT/Policy/Config/ServiceControlManager/SvchostProcessMitigation	REG_SZ
./Device/Vendor/MSFT/Policy/Config/Start/HideChangeAccountSettings	REG_SZ
./Device/Vendor/MSFT/Policy/Config/Start/HideHibernate	REG_SZ
./Device/Vendor/MSFT/Policy/Config/Start/HideLock	REG_SZ
./Device/Vendor/MSFT/Policy/Config/Start/HidePowerButton	REG_SZ
./Device/Vendor/MSFT/Policy/Config/Start/HideRestart	REG_SZ
./Device/Vendor/MSFT/Policy/Config/Start/HideShutDown	REG_SZ
./Device/Vendor/MSFT/Policy/Config/Start/HideSignOut	REG_SZ
./Device/Vendor/MSFT/Policy/Config/Start/HideSleep	REG_SZ
./Device/Vendor/MSFT/Policy/Config/Start/HideSwitchAccount	REG_SZ
./Device/Vendor/MSFT/Policy/Config/Start/HideUserTile	REG_SZ
./Device/Vendor/MSFT/Policy/Config/Start/ImportEdgeAssets	REG_SZ
./Device/Vendor/MSFT/Policy/Config/Update/ManagePreviewBuilds	REG_SZ
./Device/Vendor/MSFT/Uefi/Identity/Apply	REG_SZ
./Device/Vendor/MSFT/Uefi/Identity2/Apply	REG_SZ
./Device/Vendor/MSFT/Uefi/Permissions/Apply	REG_SZ
./Device/Vendor/MSFT/Uefi/Permissions2/Apply	REG_SZ
./Device/Vendor/MSFT/Uefi/Settings/Apply	REG_SZ
./Device/Vendor/MSFT/Uefi/Settings2/Apply	REG_SZ
./Device/Vendor/MSFT/WindowsDefenderApplicationGuard/InstallWindowsDefenderApplicationGuard	REG_SZ

User Impact

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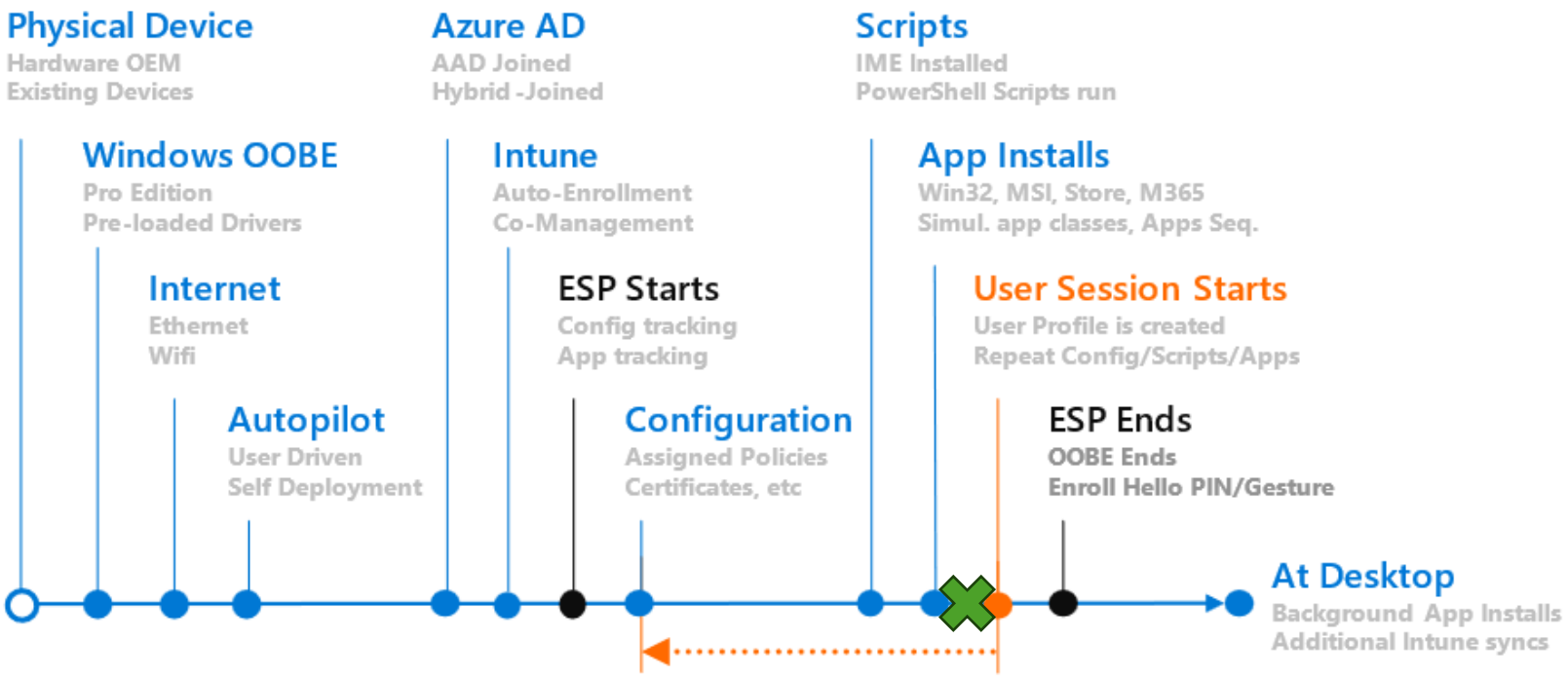
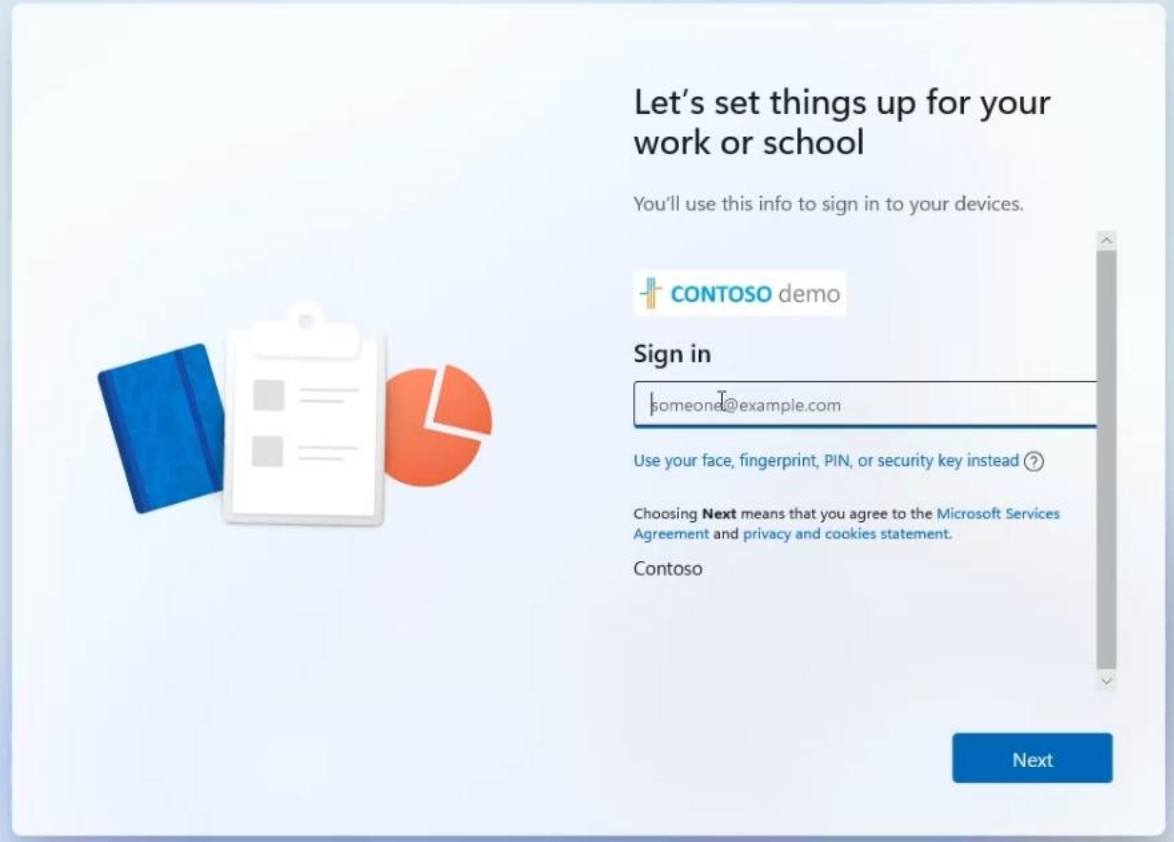
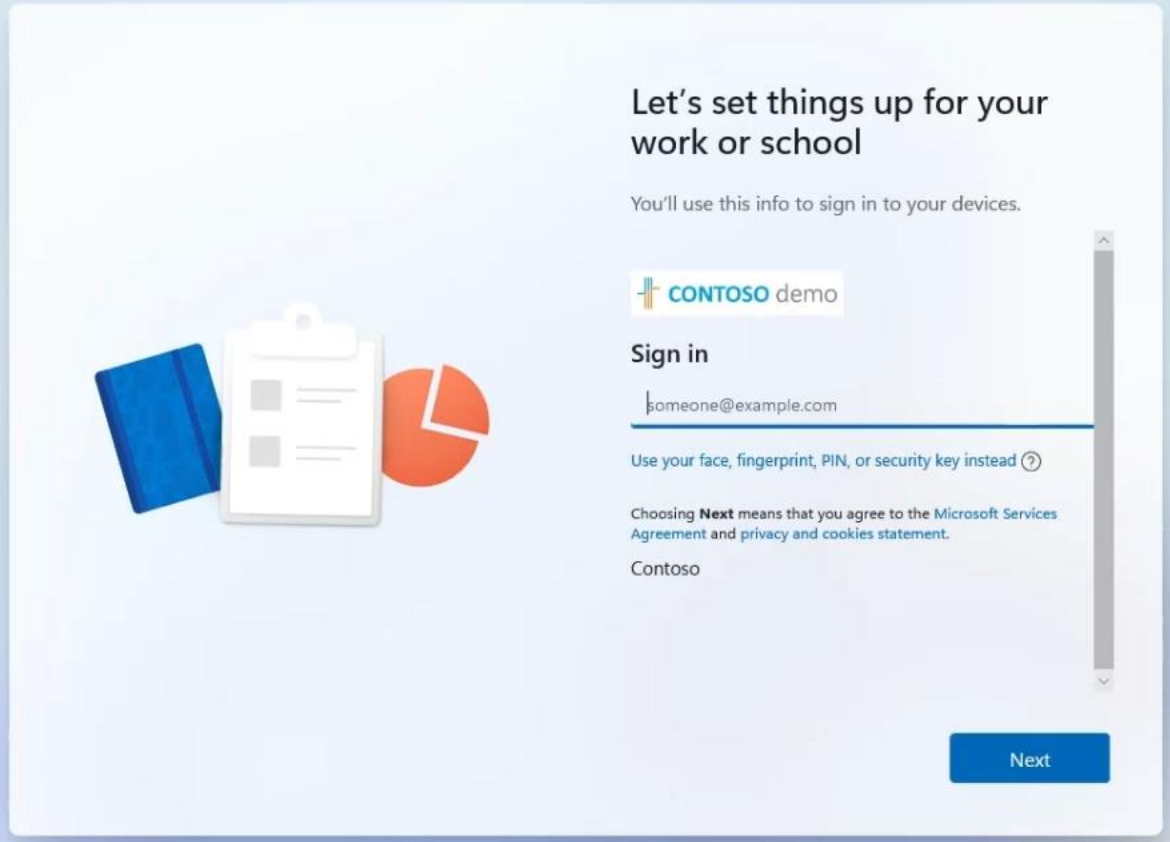
Autopilot Provisioning Sequence for AAD Join



User


Let's set things up for your work or school
You'll use this info to sign in to your devices.
CONTOSO demo
Sign in
someone@example.com
Use your face, fingerprint, PIN, or security key instead
Choosing Next means that you agree to the Microsoft Services Agreement and privacy and cookies statement.
Contoso
Next

Device


Let's set things up for your work or school
You'll use this info to sign in to your devices.
CONTOSO demo
Sign in
someone@example.com
Use your face, fingerprint, PIN, or security key instead
Choosing Next means that you agree to the Microsoft Services Agreement and privacy and cookies statement.
Contoso
Next

Extensibility

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Policy name	Policy type
Win - OIB - ES - Encryption - U - Personal Data Encryption - v3.4	Personal Data Encryption
Win - OIB - SC - Device Security - U - Device Guard, Credential Guard and HVCI - v3.5	Settings catalog
Win - OIB - SC - Device Security - U - Power and Device Lock - v3.2	Settings catalog
Win - OIB - SC - Device Security - U - Windows Sandbox - v3.4	Settings catalog
Win - OIB - SC - Device Security - U - Windows Spotlight and Org Messages - v3.0	Settings catalog
Win - OIB - SC - Microsoft Edge - U - Extensions - v3.1	Settings catalog
Win - OIB - SC - Microsoft Edge - U - Password Management - v3.0	Settings catalog
Win - OIB - SC - Microsoft Edge - U - Profiles, Sign-In and Sync - v3.0	Settings catalog
Win - OIB - SC - Microsoft Edge - U - User Experience - v3.4	Settings catalog
Win - OIB - SC - Microsoft Office - U - Config and Experience - v3.6	Settings catalog
Win - OIB - SC - Microsoft Office - U - User Security - v3.6	Settings catalog
Win - OIB - SC - Microsoft OneDrive - U - Configuration - v3.0	Settings catalog
Win - OIB - SC - Microsoft Store - U - Configuration - v3.3	Settings catalog
Win - OIB - SC - Windows User Experience - U - Copilot - v3.6	Settings catalog
Win - OIB - SC - Windows User Experience - U - Start Menu and Taskbar - v3.6	Settings catalog

Getting Involved

The curse of Intune assignments

Getting Involved

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openintunebaseline.com

- Raise an issue
- Start a discussion
- Suggest improvements
- Spread the word

We love Feedback

Great Session!

Okay Session!

Not so okay Session!

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Thank You